# Hospital Discharge Process Improvement

# Business Requirements Document

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# 1. Executive Summary

Hospitals are experiencing elevated 30-day readmission rates, exceeding national benchmarks. This results in financial penalties, operational inefficiencies, and reduced patient satisfaction. The Hospital Readmission Reduction Initiative aims to identify the key drivers of readmissions, improve discharge and follow-up processes, and provide actionable insights through a reporting dashboard.

# 2. Business Case

The hospital currently has a 30-day readmission rate of 18%, which is above the national benchmark of 14%. Contributing factors include inconsistent follow-up appointment scheduling, limited visibility into high-risk patients, and a lack of standardized discharge procedures. These issues increase operational costs and negatively affect patient outcomes.

## 2.1 Project Objectives

| Objective ID | Objective |
|---|---|
| OBJ-1 | Reduce 30-day readmission rate from 18% to 12% within 12 months |
| OBJ-2 | Ensure 90% of discharged patients have follow-up appointments scheduled |
| OBJ-3 | Provide leadership with real-time dashboards to monitor readmission trends |

# 3. Project Scope

## 3.1 In Scope

- Analysis of the current hospital discharge process
- Identification of patient readmission risk factors
- Implementation of a standardized discharge risk assessment
- Coordination of follow-up appointments prior to discharge
- Identification of barriers to medication access (transportation, delivery options)

### 3.2 Out of Scope

- Changes to physician treatment decisions
- Insurance policy or coverage modifications
- Hospital billing or financial operations
- Development of new clinical treatment protocols
- Hospital-wide IT infrastructure upgrades unrelated to readmission analysis

## 4. Stakeholder Analysis

### 4.1 Stakeholder Identification

| Stakeholder ID | Stakeholder | Department | Role |
|---|---|---|---|
| STK-01 | Hospital Administration | Executive | Project Sponsor |
| STK-02 | Physicians | Clinical | Clinical Advisors |
| STK-03 | Medical Social Worker | Care Coordination | Assess social barriers and coordinate support services |
| STK-04 | Nursing Staff | Patient Care | Discharge Workflow |
| STK-05 | IT Department | Technology | Date & Reporting |
| STK-06 | Quality Improvement Team | Compliance | Performance Monitoring |
| STK-07 | Patients | Healthcare Recipients | Receive care and follow-up instructions |

### 4.2 Stakeholder Roles and Responsibilities

| Stakeholder | Role in Project |
|---|---|

| | |
|---|---|
| Hospital Administration | Project sponsors and decision-makers |
| Physicians | Provide clinical insight into discharge processes |
| Medical Social Worker | Identify patient support needs |
| Nursing Staff | Manage patient discharge and follow-up instructions |
| IT Department | Provide data access and support dashboard implementation |
| Quality Improvement Team | Monitor readmission metrics and regulatory compliance |
| Patients | Represent the end users of the discharge process; provide perspective on patient barriers, medication access, and post-discharge support |

## 4.3 Stakeholder Influence Matrix

| Stakeholder | Influence | Interest | Engagement Strategy |
|---|---|---|---|
| Hospital Administration | High | High | Regular status updates and dashboard reports |
| Physicians | Medium | High | Consult during workflow improvement discussions |
| Medical Social Worker | Medium | High | Manage social risk factors |
| Nursing Staff | Medium | High | Include in process design workshops |
| IT Department | High | Medium | Coordinate technical requirements and reporting tools |
| Quality Improvement Team | Medium | High | Collaborate on readmission metrics and compliance tracking |
| Patient | Low | High | Ensure patients understand post-charge instructions<br><br>Collect feedback on barriers such as transportation, medication access, support at home<br><br>Provide support options where needed (pharmacy delivery, home health services, caregiver coordination) |

## 4.4 Stakeholder Communication Plan

| Stakeholder | Communication Method | Frequency |
|---|---|---|

| | | |
|---|---|---|
| Hospital Administration | Executive reports / dashboard updates | Monthly |
| Physicians | Process improvement meetings | As needed |
| Medical Social Worker | Process improvement meetings | As needed |
| Nursing Staff | Workflow workshops and training | During implementation |
| IT Department | Technical planning meetings | Weekly during development |
| Quality Improvement Team | Metrics review sessions | Monthly |
| Patients | Direct education; discharge instructions; and nurse guidance | Each discharge |

# 5. Current State (AS-IS Process)

## 5.1 Description

The current discharge process begins when a physician determines that a patient is medically stable and ready to leave the hospital. Nursing staff prepare discharge instructions, provide medication guidance, and release the patient from the hospital.

Follow-up care is sometimes scheduled prior to discharge, but this process is not standardized. As a result, some patients leave without a scheduled follow-up appointment or without clear post-discharge

care coordination. This lack of consistency can contribute to avoidable hospital readmissions within 30 days.

## 5.2 AS-IS Flowchart

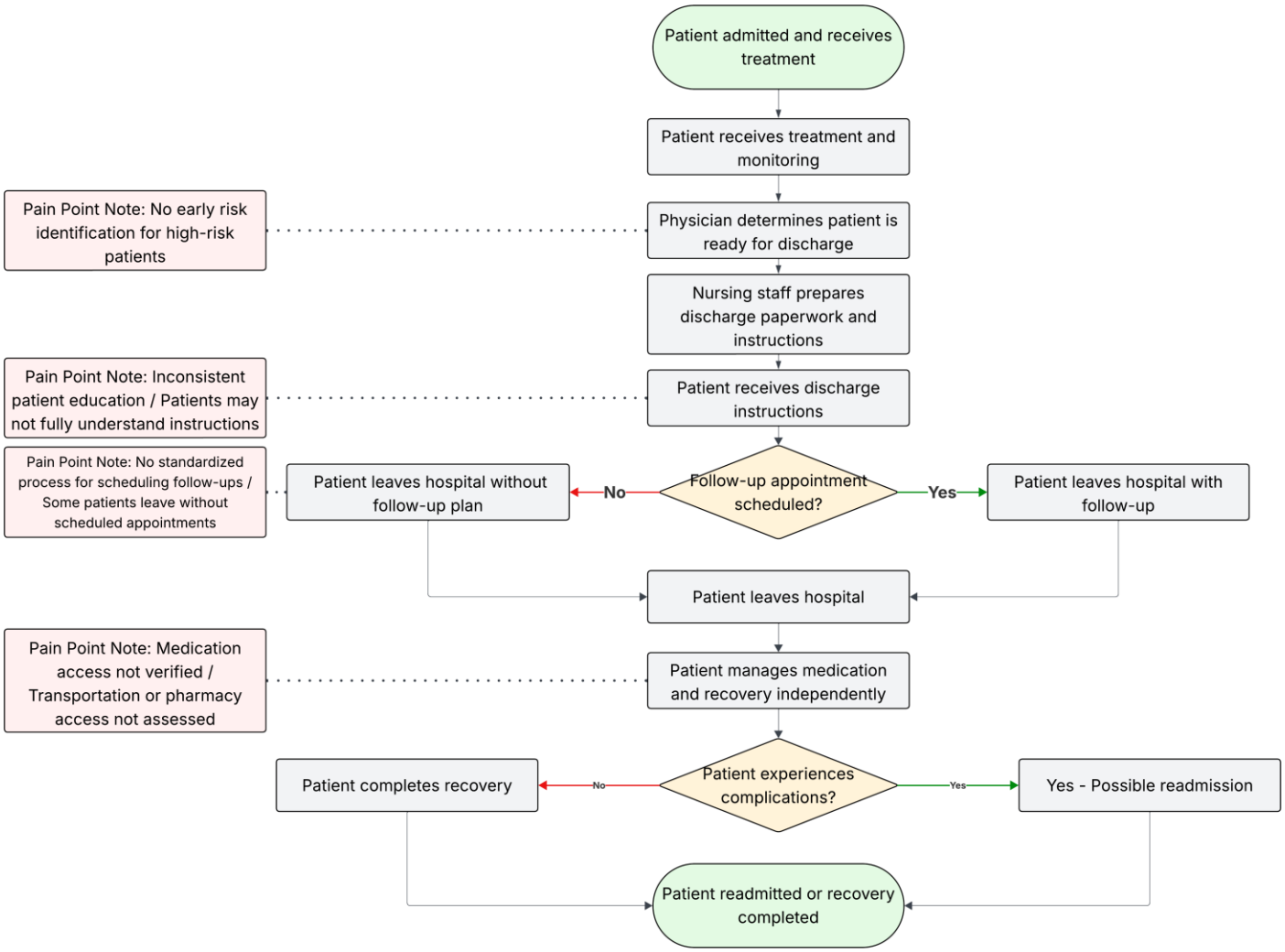


Figure 1. Current Hospital Discharge Process (AS-IS)

# 6. Pain Points and Gap Analysis

## 6.1 Pain Points

| Pain Point Identified | AS-IS Process |
|---|---|
| Inconsistent patient education | Discharge instructions vary between nursing staff, leaving patients unclear on recovery steps. |

| | |
|---|---|
| Medication access barriers | Patients may lack transportation or technical capability to obtain prescribed medications. |
| Follow-up coordination gaps | Scheduling follow-up appointments is often left to patients, leading to missed visits. |
| Delayed risk identification | High-risk patients are sometimes only flagged at discharge, limiting intervention opportunities. |

## 6.2 Gaps

- No standardized risk assessment during treatment or prior to discharge.
- Limited social worker involvement for high-risk patients.
- Inconsistent discharge communication across staff and departments.
- Insufficient use of technology for follow-up scheduling and medication delivery options.

These pain points and process gaps contribute to avoidable 30-day readmissions, inefficient resource use, and inconsistent patient experiences. Addressing them is critical to improving patient outcomes and streamlining hospital operations.

# 7. Future State (TO-BE Process)

## 7.1 Description

The improved hospital discharge process addresses the current gaps by standardizing risk assessments, involving social worker consultation for high-risk patients, and ensuring consistent patient education. Follow-up appointments are scheduled proactively before discharge, and medication access barriers are addressed through pharmacy delivery or technology-assisted ordering. These enhancements aim to reduce readmissions, improve patient outcomes, and streamline care coordination.

## 7.2 TO-BE Flowchart

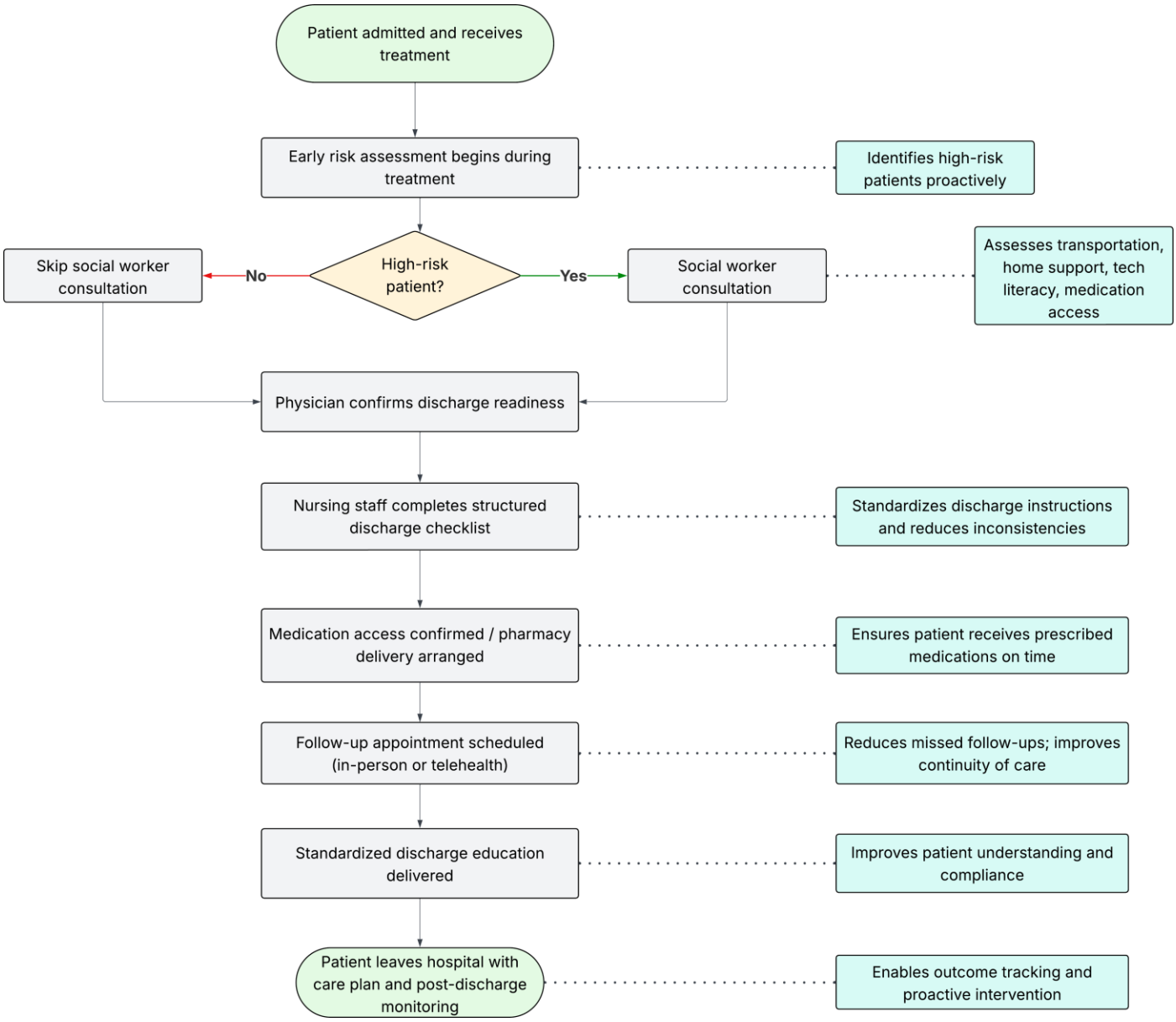


Figure 2 – Improved Hospital Discharge Process (TO-BE)

## 7.3 Addressed Pain Points

| Pain Point | TO-BE Solution |
|---|---|
| **Inconsistent patient education** | Standardized discharge instructions for all patients, ensuring clarity and consistency. |
| **Medication access barriers** | Pharmacy delivery options and technology-assisted medication ordering. |
| **Follow-up coordination gaps** | Follow-up appointments are scheduled prior to discharge with clear reminders. |
| **Delayed risk identification** | Risk assessment conducted during treatment, allowing proactive intervention for high-risk patients. |
| **Limited social worker involvement** | Automatic referral to social worker consultation for high-risk patients. |

## 7.4 Key Benefits Alignment

| TO-BE Improvement | Related Expected Benefit |
|---|---|
| Standardized patient education | Improved patient understanding of care instructions |
| Pharmacy delivery & technology-assisted medication ordering | Better access to medications |
| Pre-scheduled follow-up appointments with reminders | Improved follow-up compliance |
| Early risk assessment during treatment | Reduced hospital readmissions |
| Automatic referral to social worker for high-risk patients | Enhanced care coordination |

# 8. Expected Benefits

The improved process promotes efficiency and ensures the patient is being discharged home with clear instructions that they understand and have a plan for continued recovery at home.

## 8.1 Operational and Patient Care Benefits

| Benefit Category | Expected Benefit | Measurement / KPI |
|---|---|---|
| Patient Outcomes | Reduction in 30-day hospital readmissions through early risk identification and targeted interventions | 30-day readmission rate |
| Operational Efficiency | Standardized discharge workflow reduces inconsistencies across departments | Average discharge processing time |

| | | |
|---|---|---|
| Care Coordination | Improved collaboration between physicians, nursing staff, social workers, and pharmacy | Number of coordinated discharge interventions |
| Medication Adherence | Medication delivery coordination reduces missed or delayed prescriptions | Prescription pickup or delivery completion rate |
| Access to Follow-Up Care | Telehealth follow-up options improve appointment attendance for patients with transportation or mobility barriers | Follow-up appointment attendance rate |
| Patient Experience | Clear discharge instructions and education increase patient confidence in post-discharge care | Patient satisfaction survey results |
| Data Visibility | Dashboard reporting enables leadership to monitor readmission trends and evaluate interventions | Readmission analytics dashboard usage |

## 8.2 Additional Strategic Benefits

- Improved patient recovery outcomes
- Reduced avoidable hospital costs associated with readmissions
- Better compliance with quality care standards
- Increased hospital reputation and patient trust
- Improved data-driven decision making for leadership

# 9. Functional Requirements

| ID | Requirement | Description | Associated TO-BE Step | Priority | Responsible Stakeholder |
|---|---|---|---|---|---|
| FR-01 | Early Risk Assessment | System flags high-risk patients during treatment and alerts care team | Early risk assessment begins during treatment | High | Physician, Nurse |
| FR-02 | Social Worker Access | Social worker can view patient risk info and document interventions | Social worker consultation | High | Social Worker |
| FR-03 | Standardized Discharge Checklist | Nurses follow structured checklist for discharge | Nursing staff completes checklist | Medium | Nurse |
| FR-04 | Medication Coordination | System verifies patient can access medications and coordinates pharmacy delivery if needed | Medication access confirmed / pharmacy delivery arranged | High | Nurse, Pharmacy |

| | | | | | |
|---|---|---|---|---|---|
| FR-05 | Follow-Up Scheduling | Schedule follow-up appointments (in-person or telehealth) for all patients | Follow-up appointment scheduled | High | Nurse, Scheduling Staff |
| FR-06 | Discharge Education | Provide standardized discharge instructions, including medication and care plan | Standardized discharge education delivered | Medium | Nurse |
| FR-07 | Patient Communication | Notify patients of follow-up appointments and medication delivery | Follow-up appointment scheduled / medication access confirmed | Medium | Nurse, Social Worker |
| FR-08 | Outcome Tracking | Track post-discharge outcomes and readmission risk | End – patient leaves hospital with care plan & monitoring | Medium | Physician, Quality Team |
| FR-09 | Decision Support Alerts | Notify care team of any patients missing critical discharge steps | Nursing staff completes checklist / medication coordination | High | Nurse, Physician |
| FR-10 | Documentation Logging | Record all interventions, risk assessments, and follow-up scheduling in system | Record all interventions, risk assessments, and follow-up scheduling in system<br>All TO-BE steps | High | Nurse, Social Worker, Physician |

# 10. Use Cases

The following use cases describe how stakeholders interact with the improved discharge process to support patient care coordination and reduce readmission risk.

## 10.1 UC-01 Identify High-Risk Patients

| Field | Description |
|---|---|
| Use Case ID | UC-01 |
| Use Case Name | Identify High Risk Patients |
| Primary Actor | Physician / Nurse |
| Secondary Actors | Hospital system |
| Description | Patient risk factors are evaluated during treatment to identify potential readmission risk |
| Preconditions | Patient admitted and receiving treatment |
| Postconditions | Patient risk status recorded in system |
| Main Flow | 1. Provider reviews patient health indicators<br>2. Provider enters patient data into system<br>3. System evaluates risk criteria |

| | |
|---|---|
| | 4. System flags patient as high or low risk |
| Alternate Flow | Risk criteria unavailable → manual review performed |

## 10.2 UC-02 Social Worker Intervention

| Field | Description |
|---|---|
| Use Case ID | UC-02 |
| Use Case Name | Social Worker Risk Assessment |
| Primary Actor | Social Worker |
| Secondary Actors | Nurse, Patient |
| Description | Social worker evaluates barriers that may impact recovery after discharge |
| Preconditions | Patient identified as high risk |
| Postconditions | Support interventions documented |
| Main Flow | 1. Social worker reviews patient risk information<br>2. Social worker meets patient<br>3. Transportation, home support, and medication access assessed<br>4. Intervention plan documented |
| Alternate Flow | Patient unavailable – follow-up scheduled |

## 10.3 UC-03 Schedule Follow-Up Appointment

| Field | Description |
|---|---|
| Use Case ID | UC-03 |
| Use Case Name | Schedule Follow-Up Care |
| Primary Actor | Nurse / Scheduling Staff |
| Secondary Actors | Patient |
| Description | Follow-up appointment arranged before patient discharge |
| Preconditions | Physician approves discharge |
| Postconditions | Appointment scheduled and patient notified |
| Main Flow | 1. Nurse accesses scheduling system<br>2. Available appointments reviewed<br>3. Patient chooses in-person or telehealth option<br>4. Appointment scheduled and confirmation provided |
| Alternate Flow | Patient unavailable for scheduled time → alternate time offered |

## 10.4 UC-04 Provide Discharge Education

| Field | Description |
|---|---|
| Use Case ID | UC-04 |
| Use Case Name | Deliver Discharge Instructions |
| Primary Actor | Nurse |
| Secondary Actors | Patient |
| Description | Patient receives standardized instructions for post-discharge care |
| Preconditions | Discharge approved and checklist completed |
| Postconditions | Patient understands care plan |
| Main Flow | 1. Nurse reviews discharge checklist<br>2. Medication instructions explained<br>3. Follow-up appointment reviewed<br>4. Patient confirms understanding |
| Alternate Flow | Patient requests clarification → additional education provided |

## 10.5 UC-05 Deliver Standardized Discharge Education

| Field | Description |
|---|---|
| Use Case ID | UC-05 |
| Use Case Name | Deliver Standardized Discharge Education |
| Primary Actor | Nurse |
| Secondary Actors | Patient |
| Description | The nurse provides standardized discharge instructions to ensure the patient understands medication use, follow-up care, and recovery requirements before leaving the hospital. |
| Preconditions | Physician has approved patient discharge, and the discharge checklist is initiated. |
| Postconditions | Patient receives discharge instructions and confirms understanding of the care plan. |
| Main Flow | 1. Nurse reviews the patient discharge checklist.<br>2. Nurse retrieves standardized discharge education materials.<br>3. Nurse explains medication instructions, including dosage and timing.<br>4. Nurse reviews follow-up appointment details with the patient.<br>5. Nurse explains warning signs or symptoms requiring medical attention.<br>6. Nurse asks the patient to confirm understanding of the care plan. |

| | |
|---|---|
| | 7. Nurse documents completion of discharge education in the system. |
| Alternate Flow | • Patient does not understand instructions<br>  ○ Nurse provides additional explanation or educational materials.<br>• Nurse confirms understanding before discharge.<br>  ○ Language barrier present<br>  ○ Interpreter services or translated materials are used. |

## 10.6 UC-06 Track Post-Discharge Outcomes

| Field | Description |
|---|---|
| Use Case ID | UC-06 |
| Use Case Name | Track Post-Discharge Outcomes |
| Primary Actor | Quality Improvement Team |
| Secondary Actors | Physician, Nurse, Hospital System |
| Description | The healthcare organization monitors patient outcomes after discharge to identify complications or potential readmission risks. |
| Preconditions | Patient has been discharged and follow-up care plan is documented. |
| Postconditions | Patient outcomes are tracked and any risk indicators are flagged for intervention. |
| Main Flow | 1. System records discharge information and follow-up appointment details.<br>2. Patient attends scheduled follow-up visit or telehealth appointment.<br>3. Provider reviews patient recovery status.<br>4. System logs patient outcomes and updates monitoring records.<br>5. Quality team reviews outcome data and identifies potential readmission risks. |

| | |
|---|---|
| Alternate Flow | 1. Missed follow-up appointment<br>2. System flags missed appointments.<br>3. Care team contacts patients to reschedule.<br>4. Complication detected during follow-up<br>5. Physician initiates additional treatment or monitoring. |

## 11. Non-Functional Requirements

| ID | Requirement Type | Requirement Description | Priority |
|---|---|---|---|
| NFR-01 | Security | Patient health information must comply with HIPAA regulations and be protected from unauthorized access | High |
| NFR-02 | Access Control | Only authorized hospital staff (physicians, nurses, social workers) may access patient discharge data | High |
| NFR-03 | System Availability | The system must be available to hospital staff during operational hours with minimal downtime | High |
| NFR-04 | Performance | Patient risk assessments and alerts must be generated within 5 seconds of data entry | Medium |
| NFR-05 | Usability | Discharge checklists and patient information screens must be simple and easy for clinical staff to complete quickly | Medium |
| NFR-06 | Data Accuracy | The system must ensure accurate recording of patient discharge information and interventions | High |
| NFR-07 | Audit Logging | All patient interactions and system updates must be logged for compliance and auditing purposes | Medium |
| NFR-08 | Reliability | The system must maintain data integrity and prevent loss of patient information | High |
| NFR-09 | Scalability | The system should support increasing numbers of patient records without performance degradation | Low |

| | | | |
|---|---|---|---|
| NFR-10 | Interoperability | The system should integrate with hospital electronic health record systems and scheduling platforms | Medium |

# 12. Success Metrics (KPI's)

| KPI | Description | Target |
|---|---|---|
| Readmission Rate | % of patients readmitted within 30 days<br>Reduce by 10–20% | Reduce by 10–20% |
| Follow-Up Compliance | % of patients attending scheduled follow-up visits | 90% |
| Medication Access Confirmation | % of patients with verified medication access before discharge | 95–100% |
| High-Risk Patient Intervention | % of high-risk patients receiving social worker intervention | 100% |
| Discharge Checklist Completion | % of discharges with completed checklist | 95% |

# 13. Risks and Assumptions

This section identifies potential risks that may impact the successful implementation of the proposed discharge process improvements, as well as key assumptions made during the analysis. Documenting these factors helps stakeholders anticipate challenges and understand conditions necessary for the proposed solution to be effective.

## 13.1 Risks

| Risk ID | Risk Description | Impact | Mitigation Strategy |
|---|---|---|---|
| R-01 | Patients may not attend scheduled follow-up appointments | Higher risk of readmission | Implement appointment reminders and telehealth options |
| R-02 | Staff may resist adopting new discharge processes | Reduced compliance with new workflow | Provide training and clear discharge checklists |
| R-03 | Patients may lack transportation or support at home | Delays in medication pickup or follow-up care | Social worker assessment to identify and address barriers |
| R-04 | Technology integration issues with existing hospital systems | Delays in implementation | Conduct system compatibility testing before rollout |
| R-05 | Incomplete documentation by staff | Gaps in patient discharge information | Use mandatory checklist completion before discharge |

## 13.2 Assumptions

| Assumption ID | Assumption Description |
|---|---|
| A-01 | Hospital staff will receive training on the new discharge workflow |
| A-02 | The hospital has an electronic health record system capable of supporting discharge documentation |
| A-03 | Social workers are available to assess high-risk patients |
| A-04 | Patients have access to either in-person or telehealth follow-up appointments |
| A-05 | Pharmacy services can support medication delivery when needed |

# 14. Requirements Traceability Matrix

| Business Objective | Functional Requirement | Use Case | TO-BE Process Step |
|---|---|---|---|
| Reduce hospital readmissions | FR-01<br>Early Risk Assessment | UC-01<br>Identify High-Risk Patient | Early risk assessment during treatment |
| Reduce hospital readmissions | FR-02<br>Social Worker Access | UC-02<br>Social Worker Risk Assessment | Social worker consultation |
| Ensure patients receive medications | FR-04<br>Medication Coordination | UC-03<br>Confirm Medication Access | Medication access confirmed / pharmacy delivery arranged |
| Improve continuity of care | FR-05<br>Follow-Up Scheduling | UC-04<br>Schedule Follow-Up Appointment | Follow-up appointment scheduled |
| Improve patient understanding of care plan | FR-06<br>Discharge Education | UC-05<br>Deliver Discharge Education | Standardized discharge education delivered |
| Monitor patient outcomes | FR-08<br>Outcome Tracking | UC-06<br>Track Post-Discharge Outcomes | Post-discharge monitoring |